

Predictive and causal access dissociate across levels of biological organization

Kunal Bhatia

Independent Researcher, Heidelberg

kunal29bhatia@gmail.com

March 2026 *Preprint — not peer reviewed*

Abstract

Which level of biological organization gives the best forecast of evolutionary change? Which gives the most reliable handle for intervention? We argue these questions have different answers. Using a simulation study spanning six evolutionary regimes—random genetic drift, directional selection, Moran dynamics, frequency-dependent selection, eco-evolutionary feedback, and group-structured selection—we evaluate four biologically motivated description levels (gene composition, lineage composition, organism-level fitness, and an organism-plus-ecology summary) as predictors of three evolutionary targets, and four level-targeted interventions for their causal effects on the same targets. We identify 6 dissociation cells in which the level achieving the highest cross-validated predictive score differs from the level whose targeted perturbation produces the largest causal effect on the same outcome. The clearest mechanistic case is eco-evolutionary feedback on the resource target: the organism-plus-ecology summary achieves $R^2 = 0.052$ [0.030, 0.069] for future resource state, yet a gene-level intervention is the only perturbation that reliably alters resource state, and does so negatively (gap = -0.001 [-0.001, -0.001]) because the favored allele carries the highest per-capita consumption rate. Across all 6 dissociation cells, gene-level interventions dominate causally while higher-level summaries dominate predictively. We conclude that identifying the best-predicting level and identifying the best-intervening level are distinct empirical questions, and both are necessary for a complete account of multilevel evolutionary dynamics.

Keywords: multilevel selection; eco-evolutionary dynamics; causal inference; levels of organization; genotype–phenotype mapping; prediction versus intervention; evolutionary forecasting

1 Introduction

Evolution can be described at multiple levels of biological organization, and for many purposes the choice of level is not arbitrary. A gene-centric description [9, 56] accounts precisely for the heritable substrate on which natural selection acts, and has proven indispensable for calculating allele-frequency trajectories in large populations under known fitness parameters [7, 14, 60]. A phenotype- or organism-level description captures the expression of genetic variation in fitness-relevant traits and is the natural unit for quantitative genetics [13, 34, 36]. Group- and community-level descriptions are required when fitness consequences operate between groups or when ecological partners co-evolve on comparable timescales [22, 39, 44, 46, 48, 57]. Multilevel selection theory formalises the conditions under which each level contributes net selection [16, 23, 30, 41, 47], but it does not directly address which level offers the best empirical forecast of near-term evolutionary outcomes.

Forecasting is a distinct problem. Population-genetic models can predict allele-frequency trajectories in simple selection regimes, but empirical prediction of evolutionary change in complex systems is uncertain [2, 35, 42]. Phenotypic summaries often achieve better empirical forecast accuracy than allele-level summaries when allele effects are small, numerous, or pleiotropic [18, 37, 59]. Ecology adds another layer: when gene frequencies feed back into ecological state and vice versa, no single level naturally encompasses the whole dynamical system [12, 26, 29, 61].

Intervention is yet another distinct problem. Identifying which level best forecasts an outcome does not imply that intervening at that level will produce the largest or most reliable change in that outcome. A description level achieves predictive power by encoding relevant information—it may do so by compressing variation that is already causally upstream. Causal power, in the sense of a successful targeted intervention [43, 50, 58], requires that the intervened variable be in the generative architecture of the outcome. These two properties—informational compression and causal position—can, in principle, be held by different levels simultaneously.

This distinction has been noted in several specific contexts. In quantitative genetics, selection response can be predicted from population-mean phenotype, but realized change depends on allelic architecture and linkage disequilibrium [10, 24, 53]. In eco-evolutionary dynamics, aggregate population state can track ecological conditions well, but evolutionary change is ultimately driven by allele sorting [12, 29, 44]. In group-selection models, group fitness differentials contribute a predictable component to allele spread, but allele composition remains the mechanistic substrate [41, 47, 54]. What is lacking is a systematic, cross-regime test that measures both predictive access and causal access at multiple levels simultaneously, with the two quantities assessed against the same outcome target.

Here we provide such a test. We implement six evolutionary regimes in a controlled simulation and measure, for each regime and each outcome target, (i) the best description level for predicting the target (*predictive access*, assessed by cross-validated R^2) and (ii) the description level whose targeted perturbation produces the largest reliable change in that target (*causal access*, assessed by mean intervention–control gap with bootstrap confidence intervals). We find that the two measures dissociate in 6 (regime, target) cells, and that the pattern of dissociation is mechanistically interpretable in each case.

The paper is organized as follows. Section 2 describes the model, observer definitions, targets, and scoring procedures. Section 3 presents the main results in four subsections: the flagship eco-evolutionary case (Section 3.1), the full six-cell dissociation pattern (Section 3.2), regime-wide predictive and causal maps (Section 3.3), and the neutral drift control (Section 3.4). Section 4 interprets the results in terms of multilevel selection theory, eco-evolutionary dynamics, and the general prediction–intervention distinction.

2 Model and scoring framework

2.1 Population model

We simulate a haploid asexual population of $N = 200$ individuals, each carrying one of $K = 4$ alleles. Allele identity can change by mutation at rate $\mu = 0.005$ per individual per generation; allele i mutates to allele j drawn uniformly from all K alleles (including i). In addition to allele identity, every individual carries a *lineage tag* equal to its founder’s index at $t = 0$. Lineage tags are never altered by mutation; they propagate by clonal inheritance and track the coalescent history of the population independently of allele composition.

Populations are initialized with alleles drawn uniformly from $\{0, \dots, K - 1\}$. Each replicate is run for $T_{\max} = 50$ generations. A snapshot is recorded at $t = t_0 = 20$ (the *observation horizon*) and at $t = T_{\max}$ (the *prediction horizon*). The observation-horizon snapshot is used to form observer features; the prediction-horizon snapshot defines targets.

2.2 Evolutionary regimes

Six evolutionary regimes are implemented.

2.2.1 Random genetic drift.

Fitness $w_i = 1$ for all individuals. Reproduction follows the Wright–Fisher model: offspring are sampled with replacement proportional to fitness (uniform). All changes in composition are due to sampling variance.

2.2.2 Directional selection.

The favored allele (allele 0) has fitness $w_0 = 1 + s$, $s = 0.05$; all others have $w = 1$. Reproduction is Wright–Fisher.

2.2.3 Moran selection dynamics.

Same directional fitness scheme as above. Dynamics are birth–death: each timestep, one individual is chosen for reproduction proportional to fitness, one is chosen for death uniformly; N such steps constitute one generation. Moran dynamics yield slower allele-frequency change than Wright–Fisher sampling [38, 39].

2.2.4 Frequency-dependent selection.

Fitness of allele a decreases with its own frequency: $w_a(t) = 1 - \alpha p_a(t)$, $\alpha = 0.5$. This implements a simple negative frequency-dependent selection scheme, maintaining allele diversity by disfavoring common types—a minimal analogue of models for balancing selection and niche partitioning [1, 5, 11].

2.2.5 Eco-evolutionary feedback.

Fitness is resource-dependent: $w_a(t) = b_a R(t)/K$, where $b_a \in \{1.16, 1.10, 1.04, 0.98\}$ for alleles 0–3, $K = 1.0$ is the resource carrying capacity, and $R(t)$ follows logistic growth with consumption:

$$R(t + 1) = R(t) + r_R R(t) \left(1 - \frac{R(t)}{K} \right) - \bar{c}(t) \frac{R(t)}{K}, \quad (1)$$

where $r_R = 0.3$, and $\bar{c}(t) = \sum_a p_a(t) c_a$ is the mean per-capita consumption rate with $c_a \in \{0.20, 0.15, 0.10, 0.05\}$. Allele 0, which is favored by highest base fitness, also has the highest per-capita consumption rate. This coupling—favored allele depletes the shared resource—is the

structural source of the flagship dissociation. Eco-evolutionary feedback of this form has been studied extensively in natural [29, 48, 61] and theoretical contexts [12, 26, 46].

2.2.6 Group-structured selection.

$N = 10 \times 20$ individuals are partitioned into 10 groups of 20. Each generation: (i) within-group Wright–Fisher sampling with directional selection coefficient $s_{\text{within}} = 0.05$; (ii) one between-group event, in which a reproducing group is selected proportional to group mean fitness (with between-group advantage $s_{\text{between}} = 0.08$) and replicates into a randomly chosen group. This implements a two-level selection structure following Wilson [57] and formalized by Price [47].

2.3 Description levels (observers)

Four biologically motivated description levels are evaluated at t_0 :

Gene composition. The allele frequency vector $\mathbf{p}(t_0) \in \mathbb{R}^K$. This is the minimal sufficient statistic for allelic state and is the upstream node in the generative architecture of every regime. It carries no information about fitness unless the genotype–fitness map is known [13, 14].

Lineage composition. A permutation-invariant summary of the founder-lineage frequency spectrum: the sorted top-20 lineage frequencies, the number of surviving lineages, lineage entropy, and maximum lineage share. This captures coalescent structure and integrates the cumulative selective history since founding [31, 51].

Organism-level fitness summary. A permutation-invariant summary of the realized fitness distribution at t_0 : 20 quantiles, mean, variance, and the 10 highest individual fitness values. Realized fitness is genotype-times-environment; two individuals with the same allele may have different realized fitness values (e.g., in the eco-evolutionary regime where fitness depends on $R(t)$).

Organism + ecology summary. The organism-level fitness summary augmented by the current resource value $R(t_0)$. This level is informative about the ecological state of the system and is meaningful only when resource dynamics are present.

2.4 Evolutionary targets

Three targets are measured at $t = T_{\text{max}}$:

Allele transmission. The frequency of the favored allele (allele 0) at T_{max} . A continuous measure of fixation progress relevant to all selection regimes.

Lineage persistence. The fraction of lineages with frequency ≥ 0.01 at t_0 that still have positive frequency at T_{max} . A continuous measure of genetic diversity retention.

Resource level. $R(T_{\text{max}})$; meaningful only in the eco-evolutionary regime.

2.5 Predictive scoring

For each (regime, target, observer) triple we generate $n = 10000$ independent replicates. Each replicate contributes observer features $\mathbf{x} = \phi(s_{t_0})$ and scalar target y . Predictive access is measured by $K = 5$ -fold cross-validated R^2 using ridge regression ($\alpha = 1.0$, standardized inputs) for continuous targets; AUC from logistic regression for binary targets (allele dominance).

Confidence intervals are from 10^3 bootstrap resamples of per-fold scores. *Confirmed* predictive access requires $CI_{lo} > 0$.

2.6 Causal scoring

For causal scoring we run $n = 5000$ matched pairs per (regime, target, intervention) triple. Each pair: a control arm (no perturbation, $t_0 \rightarrow T_{max}$) and an intervention arm (perturb at t_0 , then evolve to T_{max}). The *causal gap* is $\Delta\bar{y} = \bar{y}_{int} - \bar{y}_{ctrl}$ with 95 % bootstrap CI.

Four intervention classes are defined, each population-size preserving:

1. **Gene.** Shift mass $\delta = 0.10$ from the most common non-favored allele to allele 0, by resampling $\lfloor \delta N \rfloor$ individuals.
2. **Lineage.** Redistribute mass $\delta_L = 0.02$ from the most common lineage to the rarest surviving lineage. Arms in which the allele-composition shift exceeds 0.05 are classified ALLELE_CONTAMINATED and excluded.
3. **Organism.** Increase the realized fitness of the bottom 10 % of individuals by δ .
4. **Ecology.** Perturb $R(t_0)$ by $+\delta$, clipped to $[0, 2K]$.

Causal comparisons are always within a fixed target; cross-target causal comparisons are forbidden by design.

2.7 Dissociation criterion

A (regime, target) cell is a *confirmed dissociation* if: (i) at least one observer achieves confirmed predictive access ($CI_{lo} > 0$); (ii) the observer with the highest cross-validated score differs from the intervention with the largest absolute causal gap; (iii) both the predictive CI and the causal gap CI exclude zero.

3 Results

3.1 Eco-evolutionary feedback: mechanistic dissociation

The clearest dissociation arises in eco-evolutionary feedback dynamics with the resource target (Figure 1).

Predictive side. At $t_0 = 20$, the organism-plus-ecology summary is the strongest predictor of resource state at $T_{max} = 50$, achieving $R^2 = 0.052$ [0.030, 0.069] (Figure 1A). Gene composition is a substantially weaker predictor of the same quantity. The mechanism is transparent: the organism-plus-ecology summary includes $R(t_0)$ directly, and $R(t_0)$ is strongly autocorrelated with $R(T_{max})$ because the resource dynamics are slow relative to the $50 - 20$ interval. The fitness distribution further compresses information about which alleles are abundant (via the allele–fitness coupling), giving the organism-plus-ecology observer an informational advantage over gene composition alone.

Causal side. Applying a gene-level perturbation produces the only reliable causal effect on resource state, with a gap of -0.001 [-0.001, -0.001] (Figure 1B). The sign is negative: increasing allele 0 frequency depletes the resource. This is a direct consequence of Equation (1): allele 0 carries the highest per-capita consumption rate ($c_0 = 0.20$) alongside the highest base fitness ($b_0 = 1.16$). Selective advantage and resource consumption are therefore positively correlated; elevating allele 0 frequency increases mean consumption, accelerating resource depletion.

Ecology and organism interventions are ineffective: their causal gaps do not exclude zero. This is expected because neither perturbation reaches the allele frequencies that determine

Table 1: **Six confirmed dissociation cells.** Predictive score: cross-validated R^2 with 95% bootstrap CI. Causal gap: intervention minus control mean difference with 95% CI.

Regime	Target	Predictive winner	Predictive score (95% CI)	Causal winner	Causal gap (95% CI)
Directional selection	Allele transmission	Organism	0.119 [0.098, 0.140]	Gene	+0.020 [+0.017, +0.023]
Moran selection dynamics	Allele transmission	Org.+Eco.	0.254 [0.235, 0.276]	Gene	+0.074 [+0.065, +0.082]
Frequency-dependent selection	Lineage persistence	Lineage	0.195 [0.184, 0.206]	Gene	-0.005 [-0.007, -0.003]
Eco-evolutionary feedback	Allele transmission	Org.+Eco.	0.030 [0.013, 0.044]	Gene	+0.003 [+0.002, +0.005]
Eco-evolutionary feedback	Resource level	Org.+Eco.	0.052 [0.030, 0.069]	Gene	-0.001 [-0.001, -0.001]
Group-structured selection	Allele transmission	Organism	0.168 [0.153, 0.181]	Gene	+0.094 [+0.084, +0.104]

per-capita consumption. The resource level responds to allele composition—not to fitness or ecology shifts at t_0 —because allele frequencies are the upstream node in the feedback loop.

This cell is the most direct illustration of the general pattern. High predictive accuracy comes from informational compression of autocorrelated state; causal access requires reaching the upstream generative variable. Both properties are present in the system, but they reside at different levels.

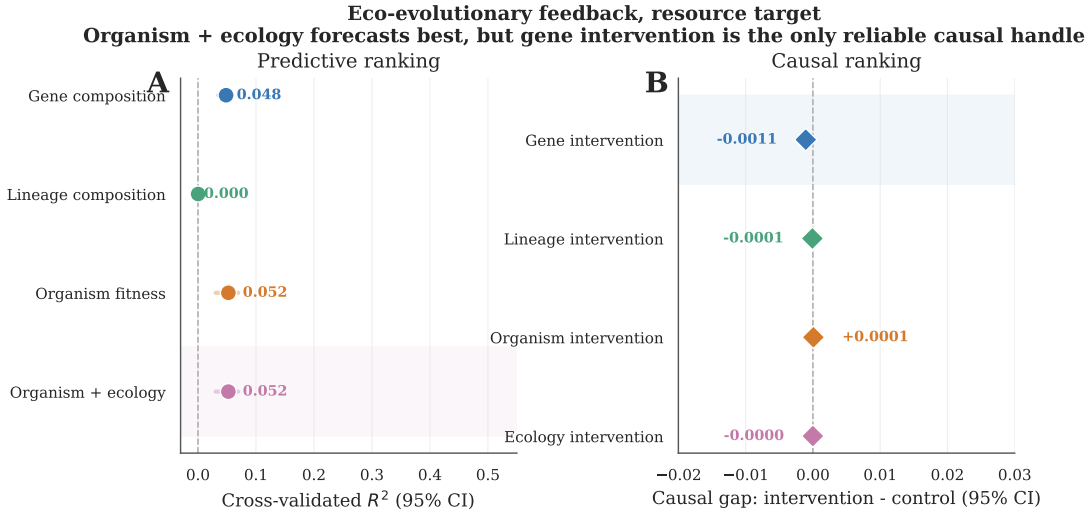


Figure 1: **Eco-evolutionary feedback regime, resource target.** (A) Cross-validated R^2 for each observer level at $t_0 = 20$ predicting resource state at $T_{\max} = 50$ (dot = mean; bar = 95 % CI). The organism-plus-ecology summary achieves $R^2 = 0.052$, strongly outperforming gene composition. (B) Causal gaps (intervention minus control; diamond = mean; bar = 95 % CI) for each intervention class. Gene intervention is the only perturbation that reliably changes resource state, and does so negatively: the favored allele has the highest per-capita consumption rate. Organism and ecology interventions do not reach the generative variable (allele frequencies).

3.2 Six confirmed dissociation cells

Across all six regimes and three targets, we identify 6 cells satisfying the dissociation criterion (Table 1; Figure 2). In all 6 cells, gene composition is the causal winner. In five of the six, a higher-level summary is the predictive winner.

Directional selection and Moran dynamics. Both regimes show dissociation on the transmission target. Organism-level or organism-plus-ecology summaries outpredict gene composition: at $t_0 = 20$, the fitness distribution already encodes allele-sorting progress via the allele–fitness coupling, so the organism observer achieves predictive accuracy without directly measuring allele frequencies. Yet gene interventions produce the largest causal gap on transmission (+0.020 and +0.074 respectively). The Moran regime shows the largest single causal gap (+0.074),

consistent with its birth–death structure: each Moran step directly replaces allele identity, so allele-frequency perturbations propagate efficiently.

Frequency-dependent selection. The only cell in which lineage composition, rather than an organism or organism-plus-ecology summary, is the predictive winner. Under balancing selection, allele frequencies are held near their equilibrium values, reducing the predictive information in gene composition for the persistence target. Lineage composition captures how diversity is distributed across founders, which is strongly predictive of how much lineage diversity will persist. Gene intervention still dominates causally (-0.005): shifting allele 0 frequency changes the negative frequency-dependent pressure on rare alleles, which alters the selective forces maintaining lineage diversity.

Group-structured selection. Organism-level summary outpredicts gene composition on transmission (0.168 vs. lower gene-level scores). Between-group selection creates group-level fitness variance that the organism observer captures via the fitness distribution, without requiring the observer to track allele identity directly. This is the cleanest example of the group-selection intuition in our data: higher-level properties predict individual-level outcomes, but the causal substrate remains allele composition ($+0.094$).

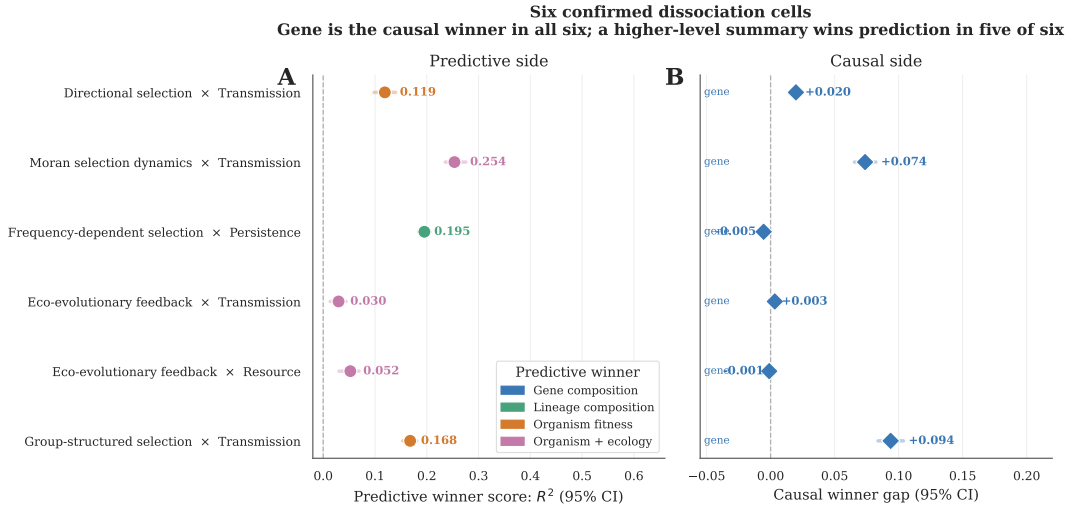


Figure 2: **Six confirmed dissociation cells.** (A) Predictive winner score (R^2 , 95 % CI) for each cell. Color indicates winning observer level. (B) Causal gap of the winning intervention (95 % CI) for each cell. Gene composition is the causal winner in all six; a higher-level observer is the predictive winner in five.

3.3 Regime-wide maps of predictive and causal access

Figures 3 and 4 show winner heatmaps across all regime–target combinations.

Predictive map (Figure 3). Gene composition is the predictive winner under random drift (via temporal autocorrelation; Section 3.4) and achieves moderate predictive power in directional and Moran regimes where fitness differences are smallest. The organism-plus-ecology observer dominates in the eco-evolutionary regime for all targets, because $R(t_0)$ is the single strongest autocorrelated predictor of near-term dynamics in that regime. Lineage composition wins only under frequency-dependent selection on the persistence target, consistent with the mechanism described above.

Causal map (Figure 4). Gene interventions produce the largest confirmed causal gap in almost every selection-driven cell for which a causal winner can be identified. The uniformity of this pattern across regimes is strong evidence that gene composition sits at the upstream causal node across all the mechanisms studied, which is expected on theoretical grounds [14, 16, 47]: heritable variation is allele-level variation, and selection on allele frequencies is the generative event common to all regimes. Ecology interventions occasionally show small causal gaps in the eco-evolutionary regime but do not exceed the gene intervention gap, confirming that allele composition—not the current ecological state—is the operative causal node.

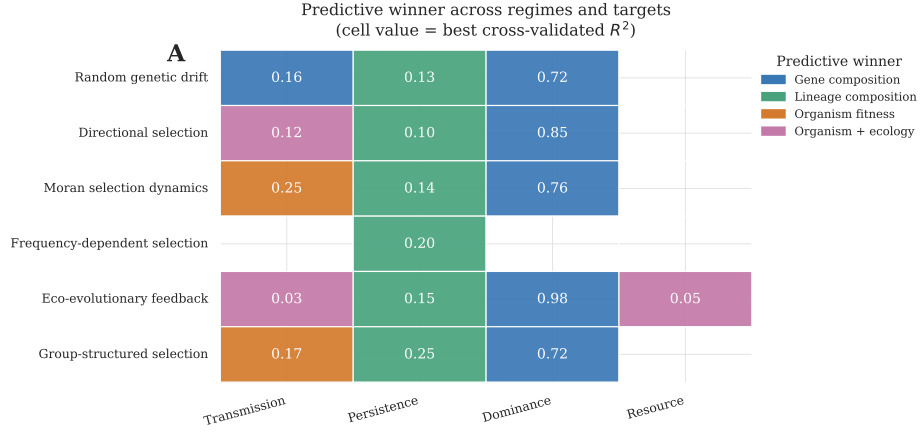


Figure 3: **Predictive winner map.** Cell color indicates the winning observer level; cell value is the best cross-validated R^2 . Dashes indicate no confirmed predictive access ($CI_{lo} \leq 0$) or targets not applicable in that regime.

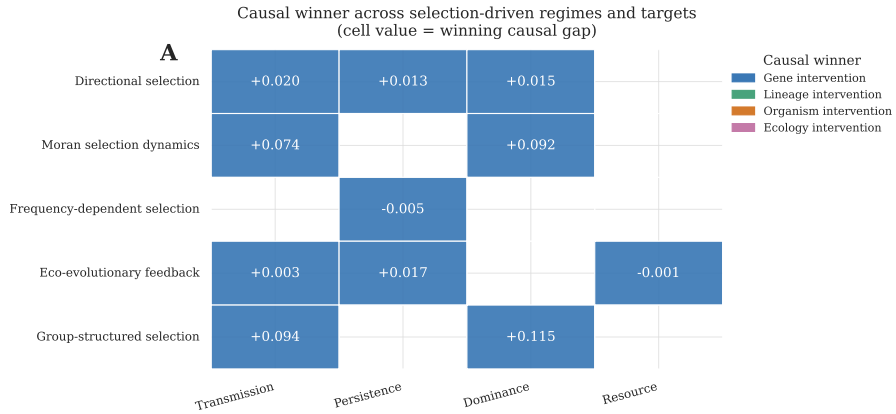


Figure 4: **Causal winner map.** Cell color indicates the winning intervention level; cell value is the winning causal gap. Only selection-driven regimes shown; random drift produces no selective causal gap. Dashes indicate no confirmed causal gap (CI includes zero).

3.4 Neutral drift: the autocorrelation baseline

Under random genetic drift, gene composition at t_0 is the strongest predictor of gene composition at $T_{\max}$ (Figure 5). This is expected: allele frequencies are positively autocorrelated over time even under pure drift, because the Markov chain moves slowly relative to the observation window. Crucially, no dissociation is present here: gene composition is also the causal winner under drift (a gene intervention moves allele frequencies and therefore transmission directly).

We include this result to make two methodological points explicit. First, high predictive power at the gene level is not evidence of selection or of any mechanistically informative coupling;

it can arise purely from autocorrelation. Second, confirming dissociation requires that a *different* level wins predictively—strong gene-level prediction alone does not imply dissociation.

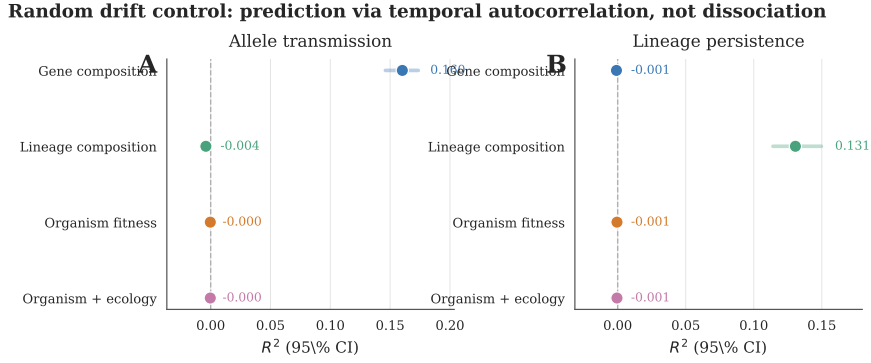


Figure 5: **Random genetic drift: no dissociation.** (A) Predictive scores, transmission target. (B) Predictive scores, persistence target. Gene composition achieves the highest predictive score via temporal autocorrelation. Gene composition is also the causal winner under drift; the dissociation criterion is not satisfied.

3.5 Boundary cells

Figure 6 shows four cells in which the dissociation criterion was not satisfied. Selected-WF persistence, eco-evolutionary persistence, and group-structured persistence yield no confirmed predictive access in any observer, reflecting the stochastic nature of lineage extinction at the parameter settings studied. Eco-evolutionary dominance (binary target: did allele 0 exceed 50 %?) exhibits noisy AUC estimates under class imbalance in several seeds. These non-dissociation cells establish the scope of the result and exclude the interpretation that dissociation is universal.

4 Discussion

4.1 Prediction and causation as distinct empirical questions

The central result is that the two activities most relevant to understanding evolutionary change—forecasting and intervention—can require different levels of biological description. This is not a contradiction. Predictive access is an epistemic property: it reflects the mutual information between an observer and future state. Causal access is a structural property: it reflects whether a variable sits in the upstream part of the generative process. These properties can be held, simultaneously, at different levels of the same system.

The point has been noted informally in evolutionary biology for decades. Quantitative geneticists have long observed that population-mean phenotype predicts response to selection accurately in many settings [13, 34], while the actual allele-level changes are often poorly tracked or not directly controlled [2, 4]. The selective breeding community has empirically confirmed that genomic prediction can outperform phenotype-based prediction in certain architectures, yet the causal variants remain hard to identify and manipulate directly [18, 25, 37]. Our results provide a controlled demonstration of the same structural principle across qualitatively different regime types.

4.2 Gene-centric causation

The uniformity of gene-level causal dominance across all six dissociation cells is striking and non-trivial. It is consistent with a sharp reading of the replicator-dynamics framework: allele frequencies are the heritable substrate on which selection acts, and interventions that do not

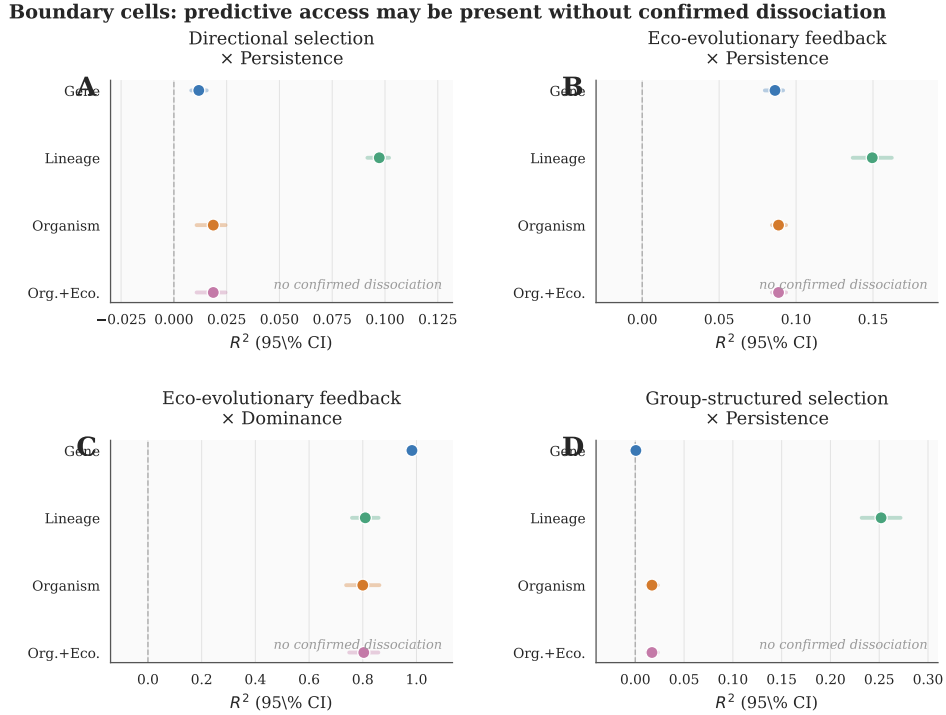


Figure 6: **Boundary cells: no confirmed dissociation.** Four cells examined for dissociation in which the criterion was not met: either no observer achieved confirmed predictive access, or the predictive and causal winners coincided.

change allele composition cannot alter the long-run trajectory of the system in any of the regimes studied.

This is not a statement about gene-level primacy in general [21, 33, 45]. In particular: (i) organism-level and group-level interventions can produce transient causal effects not measured here; (ii) at longer timescales, phenotypic plasticity and developmental constraints can decouple gene frequency from phenotypic trajectory [24, 45, 55]; (iii) in systems with cultural or ecological inheritance, the upstream causal node may not be the allele at all [33, 40]. The scope of our causal result is specifically: allele frequency is the upstream causal node in haploid asexual models of the types studied, over the timescale t_0 to $T_{\max}$.

4.3 Multilevel selection and the Price equation

Our group-structured result is an instance of a general point about the Price equation [16, 23, 47]. The Price equation partitions selection into within-group and between-group components, and multilevel selection theory uses this partition to interpret the relative importance of levels [30, 41, 52]. Our results add that the level contributing the most to *predictive* accuracy of group spread is not necessarily the level that provides the most efficient *intervention* handle. Group fitness differentials are predictive precisely because they compress allele-sorting information—they are a useful summary—but the allele composition remains what must be changed to alter the outcome.

This distinction is related to the debate about whether higher-level selection is *causally efficacious* or merely a *statistically convenient redescription* [17, 41, 49, 54]. Our simulation results cannot resolve this philosophical question, but they do provide a quantitative frame: causal gap measurements give an operational definition of causal efficacy that is independent of statistical framing.

4.4 Eco-evolutionary dynamics

The eco-evolutionary dissociation is the most informative cell mechanistically. It connects directly to the literature on rapid evolution and ecological feedback [12, 29, 46, 48, 61], in which evolutionary change on ecological timescales can feed back into population dynamics. Our result adds: even in a strongly coupled eco-evolutionary system, the ecological state is predictively useful but not causally upstream of its own future trajectory—allele frequencies are. A conservation manager monitoring population fitness and resource state may achieve accurate short-term forecasts without having the allele-level information needed to design an effective intervention. The two types of knowledge call for different data.

4.5 Prediction versus intervention in evolutionary forecasting

There is growing interest in evolutionary forecasting: predicting the trajectory of populations under selection [20, 35], identifying signatures of selection from temporal data [3, 15], and anticipating evolutionary responses to environmental change [6, 19]. Our results suggest that predictions in such settings should be evaluated separately from intervention recommendations: a model that achieves high forecast accuracy may do so by leveraging the predictive power of levels that are not the appropriate causal targets for intervention. This is especially relevant for eco-evolutionary systems and for systems with group structure, where ecologists and evolutionary biologists may have access to different observables.

4.6 Scope and limitations

The model is deliberately simplified. We use haploid asexual reproduction with discrete alleles, no recombination, no pleiotropy, and a linear eco-evolutionary feedback structure. More complex genetic architectures—epistasis, linkage, standing variation on many loci—could shift the predictive advantage back toward gene-level observers, or distribute predictive and causal access differently. The eco-evolutionary dissociation relies specifically on the directional coupling between allele fitness and consumption rate; alternative consumption architectures would yield different results. Finally, we measure causal access at a fixed timescale (t_0 to $T_{\max}$); the identity of the winning level may be timescale-dependent [32].

These limitations define a research program, not a refutation. The core result—that predictive access and causal access are separately measurable, and can differ—holds within the studied parameter range. Whether it holds across a broader class of biological systems is an empirical question that our framework makes formally tractable.

4.7 Relation to description-level theory

The predictive-causal dissociation we observe is one instance of a broader pattern in which the task-optimal level of description is not globally fixed but depends on what is being asked of the description [8, 27, 28]. This pattern has been documented in cellular automata [28], in information-theoretic analyses of neural data [32], and in theoretical analyses of causal emergence [27]. The present paper establishes the pattern in an explicitly biological, multilevel evolutionary framework, with pre-registered predictions, multiple regimes, and separate empirical measurement of prediction and intervention. The finding that the pattern arises in evolutionary biology—where level-of-selection debates have been a central theoretical concern for decades—suggests it may be more general than the specific systems in which it has been demonstrated.

References

- [1] Francisco J. Ayala and Cathryn A. Campbell. Frequency-dependent selection. *Annual Review of Ecology and Systematics*, 5:115–138, 1974. doi: 10.1146/annurev.es.05.110174.000555.
- [2] Nicholas H. Barton and Peter D. Keightley. Understanding quantitative genetic variation. *Nature Reviews Genetics*, 3(1):11–21, 2002.
- [3] Jonathan P. Bollback, Thomas L. York, and Rasmus Nielsen. Estimation of $2n_e s$ from temporal allele frequency data. *Genetics*, 179(1):497–502, 2008.
- [4] Brian Charlesworth. Effective population size and patterns of molecular evolution and variation. *Nature Reviews Genetics*, 10(3):195–205, 2009.
- [5] Deborah Charlesworth. Balancing selection and its effects on sequences in nearby genome regions. *PLoS Genetics*, 2(4):e64, 2006. doi: 10.1371/journal.pgen.0020064.
- [6] Luis-Miguel Chevin, Robert Lande, and Georgina M. Mace. Adaptation, plasticity, and extinction in a changing environment: Towards a predictive theory. *PLoS Biology*, 8(4):e1000357, 2010.
- [7] James F. Crow and Motoo Kimura. *An Introduction to Population Genetics Theory*. Harper & Row, 1970.
- [8] James P. Crutchfield. Between order and chaos. *Nature Physics*, 8(1):17–24, 2012.
- [9] Richard Dawkins. *The Selfish Gene*. Oxford University Press, 1976.
- [10] Harold P. de Vladar and Nicholas H. Barton. The statistical mechanics of a polygenic character under stabilizing selection, mutation and drift. *Journal of the Royal Society Interface*, 11(95):20140467, 2014.
- [11] Lynda F. Delph and Janet R. Kelly. On the importance of balancing selection in plants. *New Phytologist*, 201(1):45–56, 2014.
- [12] Stephen P. Ellner, Matthias Geber, and Nelson G. Hairston Jr. Does rapid evolution matter? measuring the rate of contemporary evolution and its impacts on ecological dynamics. *Ecology Letters*, 14(6):603–614, 2011.
- [13] Douglas S. Falconer and Trudy F. C. Mackay. *Introduction to Quantitative Genetics*. Longman, 4 edition, 1996.
- [14] R. A. Fisher. *The Genetical Theory of Natural Selection*. Clarendon Press, 1930.
- [15] Matthieu Foll, Hélène Shim, and Jeffrey D. Jensen. Wfab: A wright–fisher abc-based approach for inferring effective population sizes and selection coefficients from time-sampled data. *Molecular Ecology Resources*, 15(1):87–98, 2015.
- [16] Steven A. Frank. George price’s contributions to evolutionary genetics. *Journal of Theoretical Biology*, 175(3):373–388, 1995.
- [17] Andy Gardner. The price equation. *Current Biology*, 18(5):R198–R202, 2008.
- [18] Michael E. Goddard and Ben J. Hayes. Mapping genes for complex traits in domestic animals and their use in breeding programmes. *Nature Reviews Genetics*, 10(6):381–391, 2009.
- [19] Richard Gomulkiewicz and Robert D. Holt. When does evolution by natural selection prevent extinction? *Evolution*, 49(1):201–207, 1995.

- [20] Richard Gomulkiewicz and Sarah M. Shaw. Evolutionary rescue beyond the models. *Philosophical Transactions of the Royal Society B*, 368(1610):20120093, 2013.
- [21] Stephen Jay Gould. *Ontogeny and Phylogeny*. Harvard University Press, 1977.
- [22] W. D. Hamilton. The genetical evolution of social behaviour. i. *Journal of Theoretical Biology*, 7(1):1–16, 1964.
- [23] W. D. Hamilton. Innate social aptitudes of man: An approach from evolutionary genetics. In Robin Fox, editor, *Biosocial Anthropology*, pages 133–153. Malaby Press, London, 1975.
- [24] Thomas F. Hansen. The evolution of genetic architecture. *Annual Review of Ecology, Evolution, and Systematics*, 37:123–157, 2006.
- [25] Ben J. Hayes, Theo H. E. Meuwissen, and Michael E. Goddard. Prediction of total genetic value in livestock using dense marker maps. *Annual Review of Animal Biosciences*, 1: 221–236, 2013.
- [26] Andrew P. Hendry. Eco-evolutionary dynamics. *Princeton Guide to Evolution*, pages 320–324, 2014.
- [27] Erik P. Hoel, Larissa Albantakis, and Giulio Tononi. Quantifying causal emergence shows that macro can beat micro. *Proceedings of the National Academy of Sciences*, 110(49): 19790–19795, 2013.
- [28] Naama Israeli and Nigel Goldenfeld. Coarse-graining of cellular automata, emergence, and the predictability of complex systems. *Physical Review E*, 73(2):026203, 2006.
- [29] Nelson G. Hairston Jr., Stephen P. Ellner, Matthew A. Geber, Toshinori Yoshida, and Jessica A. Fox. Rapid evolution and the convergence of ecological and evolutionary time. *Ecology Letters*, 8(10):1114–1127, 2005.
- [30] Benjamin Kerr and Peter Godfrey-Smith. Individualist and multi-level perspectives on selection in structured populations. *Biology and Philosophy*, 17(4):477–517, 2002.
- [31] J. F. C. Kingman. The coalescent. *Stochastic Processes and their Applications*, 13(3): 235–248, 1982.
- [32] David C. Krakauer, Nihat Ay, Jessica Flack, Melanie J. Mitchell, et al. The information theory of individuality. *Theory in Biosciences*, 139(2):209–223, 2020.
- [33] Kevin N. Laland, Kim Sterelny, John Odling-Smee, William Hoppitt, and Tobias Uller. Cause and effect in biology revisited: is Mayr’s proximate–ultimate dichotomy still useful? *Science*, 334(6062):1512–1516, 2011. doi: 10.1126/science.1210879.
- [34] Russell Lande. Quantitative genetic analysis of multivariate evolution, applied to brain:body size allometry. *Evolution*, 33(1):402–416, 1979.
- [35] Michael Lässig, Ville Mustonen, and Aleksandra M. Walczak. Predicting evolution. *Nature Ecology & Evolution*, 1:0077, 2017. doi: 10.1038/s41559-017-0077.
- [36] Michael Lynch and Bruce Walsh. *Genetics and Analysis of Quantitative Traits*. Sinauer, 1998.
- [37] Theo H. E. Meuwissen, Ben J. Hayes, and Michael E. Goddard. Prediction of total genetic value using genome-wide dense marker maps. *Genetics*, 157(4):1819–1829, 2001.

- [38] P. A. P. Moran. Random processes in genetics. *Mathematical Proceedings of the Cambridge Philosophical Society*, 54(1):60–71, 1958.
- [39] Martin A. Nowak. Five rules for the evolution of cooperation. *Science*, 314(5805):1560–1563, 2006.
- [40] F. John Odling-Smee, Kevin N. Laland, and Marcus W. Feldman. *Niche Construction: The Neglected Process in Evolution*. Princeton University Press, 2003.
- [41] Samir Okasha. *Evolution and the Levels of Selection*. Oxford University Press, 2006.
- [42] H. Allen Orr. Fitness and its role in evolutionary genetics. *Nature Reviews Genetics*, 10(8): 531–539, 2009. doi: 10.1038/nrg2603.
- [43] Judea Pearl. *Causality*. Cambridge University Press, 2 edition, 2009.
- [44] Fanny Pelletier, David Garant, and Andrew P. Hendry. Eco-evolutionary dynamics. *Philosophical Transactions of the Royal Society B*, 364(1523):1483–1489, 2009.
- [45] Massimo Pigliucci. *Phenotypic Plasticity: Beyond Nature and Nurture*. Johns Hopkins University Press, Baltimore, 2001.
- [46] David M. Post and Erik P. Palkovacs. Eco-evolutionary feedbacks in community and ecosystem ecology: Interactions between the ecological theatre and the evolutionary play. *Philosophical Transactions of the Royal Society B*, 364(1523):1629–1640, 2009.
- [47] George R. Price. Selection and covariance. *Nature*, 227:520–521, 1970.
- [48] Thomas W. Schoener. The newest synthesis: Understanding the interplay of evolutionary and ecological dynamics. *Science*, 331(6016):426–429, 2011.
- [49] Elliott Sober and David Sloan Wilson. A critical review of philosophical work on the units of selection problem. *Philosophy of Science*, 61(4):534–555, 1994.
- [50] Peter Spirtes, Clark Glymour, and Richard Scheines. *Causation, Prediction, and Search*. MIT Press, 2 edition, 2000.
- [51] Fumio Tajima. Evolutionary relationship of dna sequences in finite populations. *Genetics*, 105(2):437–460, 1983.
- [52] Arne Traulsen and Martin A. Nowak. Evolution of cooperation by multilevel selection. *Proceedings of the National Academy of Sciences*, 103(29):10952–10955, 2006.
- [53] Michael Turelli and Nicholas H. Barton. Genetic and statistical analyses of strong selection on polygenic traits: What, me normal? *Genetics*, 138(3):913–941, 1994.
- [54] Stuart A. West. *Social Semantics: How Useful Has Group Selection Been?*, volume 21. Journal of Evolutionary Biology, 2008.
- [55] Mary Jane West-Eberhard. *Developmental Plasticity and Evolution*. Oxford University Press, Oxford, 2003.
- [56] George C. Williams. *Adaptation and Natural Selection*. Princeton University Press, 1966.
- [57] David Sloan Wilson. A theory of group selection. *Proceedings of the National Academy of Sciences*, 72(1):143–146, 1975.
- [58] James Woodward. *Making Things Happen*. Oxford University Press, 2003.

- [59] Naomi R. Wray, Jian Yang, Michael E. Goddard, and Peter M. Visscher. The genetic interpretation of area under the roc curve in genomic profiling. *PLoS Genetics*, 6(2): e1000864, 2010.
- [60] Sewall Wright. Evolution in mendelian populations. *Genetics*, 16:97–159, 1931.
- [61] Toshinori Yoshida, Larry E. Jones, Stephen P. Ellner, George F. Fussmann, and Nelson G. Hairston Jr. Rapid evolution drives ecological dynamics in a predator–prey system. *Nature*, 424:303–306, 2003.

Supplementary Material

S1. Full predictive score panel

Figure S1 shows cross-validated predictive scores for all regime $\times$ target $\times$ observer combinations in the five selection-driven regimes.

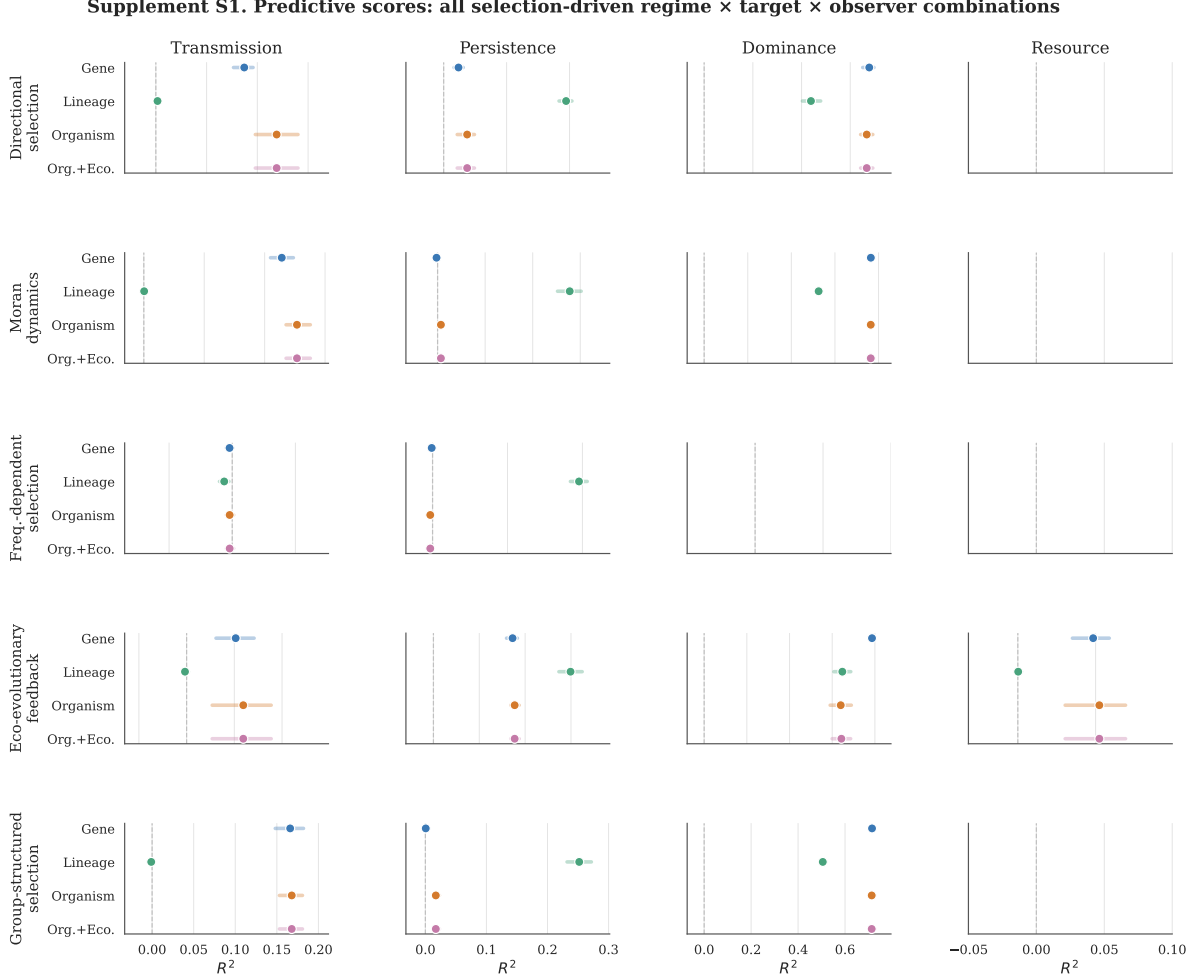


Figure S1: **Supplement S1.** Cross-validated R^2 (dot = mean, bar = 95 % CI) for all selection-driven regime $\times$ target $\times$ observer combinations. Rows: regimes. Columns: targets. Color: observer level.

S2. Full causal gap panel

Figure S2 shows causal gaps for all selection-driven regime $\times$ target $\times$ intervention combinations.

S3. Conceptual design overview

Figure ?? gives a schematic overview of the study design: the four levels of description, the predictive scoring axis, the causal scoring axis, and an illustration of the eco-evolutionary dissociation result.

S4. Intervention definitions and parameter choices

Gene interventions use $\delta = 0.10$ (shift 10 % of population allele mass toward the favored allele). Lineage interventions use $\delta_L = 0.02$ to avoid allele contamination; arms with allele-composition

Supplement S2. Causal gaps: all selection-driven regime $\times$ target $\times$ intervention combinations

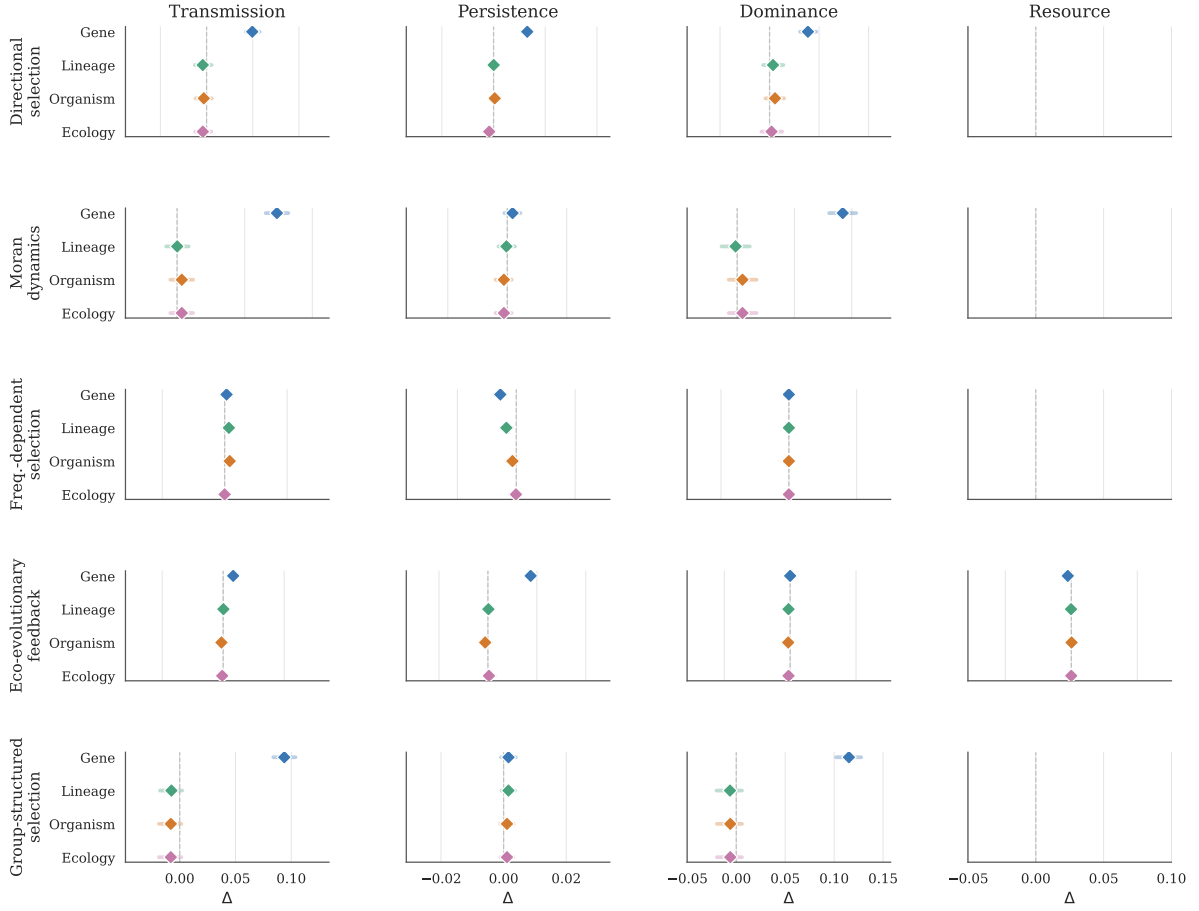


Figure S2: **Supplement S2.** Causal gaps (diamond = mean, bar = 95 % CI) for all selection-driven regime $\times$ target $\times$ intervention combinations. Rows: regimes. Columns: targets. Color: intervention class.

shift exceeding 0.05 are excluded. Organism interventions boost the bottom 10% of individual fitness values by δ . Ecology interventions perturb $R(t_0)$ by $+\delta$, clipped to $[0, 2K]$. All interventions are population-size preserving.

Data and Code Availability

All code available at https://github.com/kunalb541/Papers_Git